costing_SIGNOFF

Final sign-off (prototype stage)

Generated by `manuals/gen_signoff.py` from the release folders, the design review (`costing/REVIEW_findings.json`), the enclosure checks and `costing/CONFIDENCE.md`. **APPROVED FOR PROTOTYPING**: order a first small batch (5 boards / 1 kit). **APPROVED WITH CONDITIONS**: prototype allowed, the condition must be met before the step it names. **HOLD**: do not order.

Group	Project	Sign-off	PCB	Enclosure	Order conf.	Works conf.	Condition / first step
Utility tools	Seeti Counter	APPROVED FOR PROTOTYPING	order-ready	box v2, 2xAAA bay	Medium-High	Medium	Bench-test the microphone gain with real claps before ordering more than 5 boards.
Utility tools	Gas Taraju (LPG cylinder scale)	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAA bay	Medium-High	Medium	Calibrate with a known weight; check drift over a day.
Utility tools	Fridge Door Guard	APPROVED FOR PROTOTYPING	order-ready	box v2, 2xAAA bay	Medium-High	Medium-High	Test the reed gap and the 2-minute alarm on the bench.
Utility tools	Bijli Meter Reader	APPROVED FOR PROTOTYPING	order-ready	box v2	High	Medium	Try on a real meter with a shroud before enclosure work.
Utility tools	Bijli Log	APPROVED FOR PROTOTYPING	order-ready	box v2, 2xAAA bay	High	Medium	Do not connect to mains until a qualified electrician reviews the isolation; start with a bench supply.
Utility tools	Chulha Timers	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAAA bay	Medium-High	Medium-High	Check button debounce and the buzzer volume on the first board.
Utility tools	Rain Alert Bell + plate	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAA bay	Medium-High	Medium	Use the plate for a week outdoors before deciding on a coating.
Utility tools	Tap Litre Meter	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAA bay	Medium-High	Medium	Calibrate against a measured bucket.
Utility tools	Masala Rack (laser-cut)	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium-High	Cut a kerf coupon first.
Utility tools	Laptop Stand (laser-cut)	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium-High	Cut a kerf coupon and load-test.
Products	1 Desi Games Studio (print)	APPROVED WITH CONDITIONS	no PCB		Medium-High	Medium	native-speaker language and fact review before the production print

Group	Project	Sign-off	PCB	Enclosure	Order conf.	Works conf.	Condition / first step
Products	2 Kalakriti Automata (laser)	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium	Cut one kit in scrap MDF and run the mechanism.
Products	3 Prayog STEM lab kit	APPROVED FOR PROTOTYPING	order-ready	box v2	High	Medium-High	Run the kit's own experiments on the first unit.
Products	4 Jal-Rakshak tank controller	APPROVED WITH CONDITIONS	order-ready	box v2	Medium-High	Medium	12 V low-voltage prototype only; a mains version needs isolation design and certification
Products	5 Kheti-Mitra irrigation	APPROVED FOR PROTOTYPING	order-ready	box v2	Medium-High	Medium	Run with a lamp as the load for 48 hours before any valve or pump.
Products	6 Gas-Suraksha leak alarm	APPROVED WITH CONDITIONS	order-ready	box v2, 3xAA bay	High	Low-Medium	not a certified gas detector: data-logging prototype only, never a safety device
Toys (10)	01 Reflex Raja	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAAA bay	Medium-High	Medium-High	Play-test the buttons and the buzzer.
Toys (10)	02 Taal Memory	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAAA bay	Medium-High	Medium-High	Play-test the buttons and the buzzer.
Toys (10)	03 Aloo Bomb	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAAA bay	Medium-High	Medium-High	Test timing with a real potato.
Toys (10)	04 Beat Runner	APPROVED FOR PROTOTYPING	order-ready	box v2, 3xAAA bay	Medium-High	Medium-High	Check the LED brightness and button spacing.
Toys (10)	05 Rangoli Spiro Gears	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium-High	Cut a kerf coupon.
Toys (10)	06 Paper Rocket Launcher	APPROVED WITH CONDITIONS	no PCB		Medium-High	Medium	measure range and energy on the first unit; foam noses mandatory; toy-standard review before sale
Toys (10)	07 Tilt Maze	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium	Cut one set in scrap MDF and test the tilt range and axle fits.
Toys (10)	08 Rubber-band Racer	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium-High	Cut one in cardboard or MDF and test the band drive.
Toys (10)	09 Marble Cricket	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium	Cut one, drop 200 marbles and check the pocket distribution.

Group	Project	Sign-off	PCB	Enclosure	Order conf.	Works conf.	Condition / first step
Toys (10)	10 Katha Cubes	APPROVED WITH CONDITIONS	no PCB		Medium-High	Medium-High	native-speaker review of Hindi/Telugu before the production print
Bio / agri / civil / robotics programmes	KINETIC motion controller + vibration puck	APPROVED FOR PROTOTYPING	order-ready	box v2	High	Medium	Power with a current-limited supply and no drivers; then one driver and one motor at low current.
Bio / agri / civil / robotics programmes	MITTI-NETRA soil + structure node	APPROVED FOR PROTOTYPING	order-ready	plate for IP65 box	High	Medium	Bring up power and charging first (check TEMP voltage at room temperature), then the modem last.
Bio / agri / civil / robotics programmes	BioTray vertical-farm controller	APPROVED FOR PROTOTYPING	order-ready	plate for IP65 box	Medium-High	Medium	Dummy loads on all outputs first; seal the enclosure before any water.
Bio / agri / civil / robotics programmes	BioWatt MFC harvester	APPROVED WITH CONDITIONS	order-ready	box v2	Medium-High	Low-Medium	bench-supply test first; one microbial fuel cell is marginal for cold start (use two in series)
Bio / agri / civil / robotics programmes	GEONODE foundation monitoring	APPROVED WITH CONDITIONS	no PCB		Medium	Low-Medium	thresholds must be re-derived from a real baseline recording before any alarm is trusted
Bio / agri / civil / robotics programmes	AQUAPONICS controller	APPROVED WITH CONDITIONS	no PCB		Medium	Low-Medium	water-only trial for a week; biological limits set with an aquaculture expert before fish
Bio / agri / civil / robotics programmes	SORTER bio-robotic platform	APPROVED WITH CONDITIONS	no PCB		Medium	Low	machine guarding and an emergency stop before any powered test; vision model must be retrained on real produce
Bio / agri / civil / robotics programmes	AGROVER solar rover	APPROVED WITH CONDITIONS	no PCB		Medium	Low-Medium	hardware emergency stop and a safety review before outdoor driving
Other projects	Jal-Netra (engineering marvel)	APPROVED WITH CONDITIONS	order-ready	box v2	Medium-High	Low-Medium	run the Gate 1 acoustic test before building more than 2 boards
Other projects	ASTRADAND gesture wand (sci-fi project)	APPROVED FOR PROTOTYPING	order-ready	old enclosure	Medium	Medium	Order 2 boards, check the LED footprint fit, then run the gesture self-test.

Group	Project	Sign-off	PCB	Enclosure	Order conf.	Works conf.	Condition / first step
Other projects	Gravity toys (5)	APPROVED FOR PROTOTYPING	no PCB		Medium-High	Medium	Cut or turn one of each and play-test; measure ramp times and marble behaviour.
Other projects	TRISHNA urinal hydration monitor	APPROVED WITH CONDITIONS	order-ready	custom CAD (trishna/mechanical)	Medium-High	Low-Medium	wellness prototype only (not a medical device); leak-test the printed flow cell and pass validation V1-V4 before real use; SG must be fitted to a refractometer
Other projects	Wave 1 electronics toys	APPROVED WITH CONDITIONS	order-ready	box v2, 3xAA bay	Medium-High	Low-Medium	check the MP3 command codes and the display-module polarity on the first board; toy-safety (IS 9873) testing before sale
Other projects	Prototype packs (10)	APPROVED FOR PROTOTYPING	no PCB		Medium	Medium	Use as learning kits; do not sell.
Other projects	DJ / party system	APPROVED WITH CONDITIONS	no PCB		Medium	Low-Medium	built from bought equipment: follow the plan; mains equipment wiring by a qualified electrician

Totals: APPROVED FOR PROTOTYPING: 27, APPROVED WITH CONDITIONS: 14, HOLD: 0